

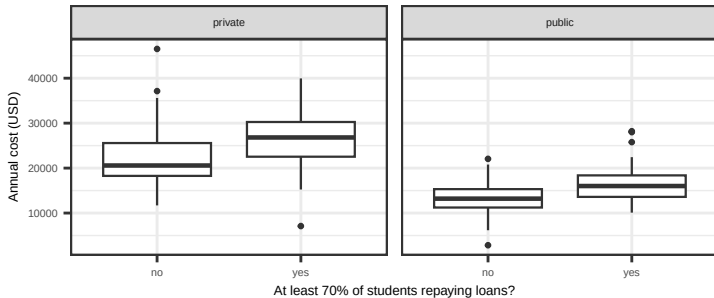
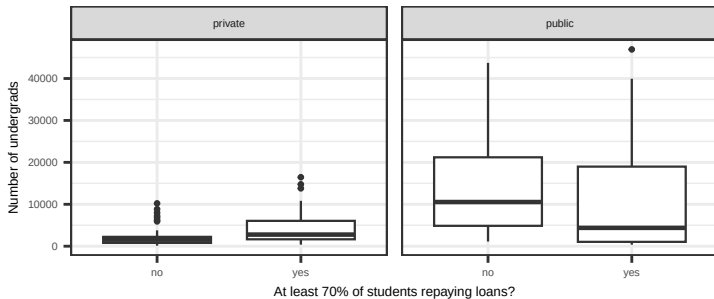
Model diagnostics

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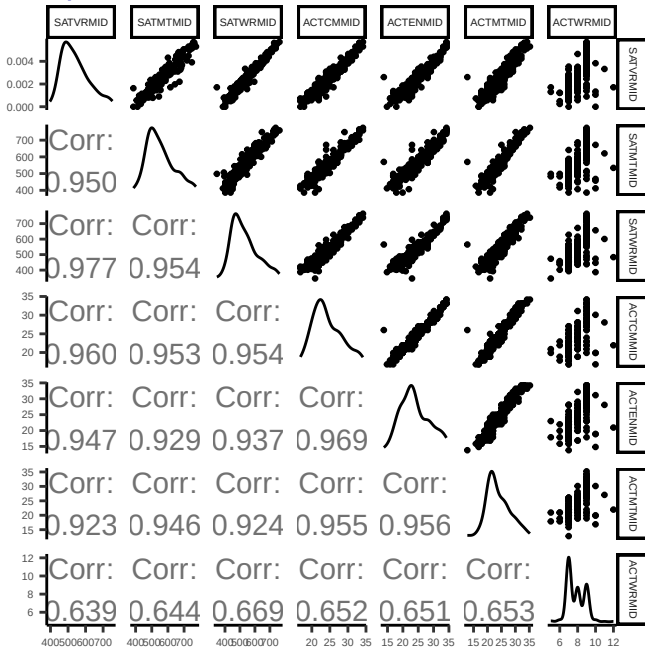
Warmup

Work on the warmup activity (handout), then we will discuss as a class.

Warmup



Warmup



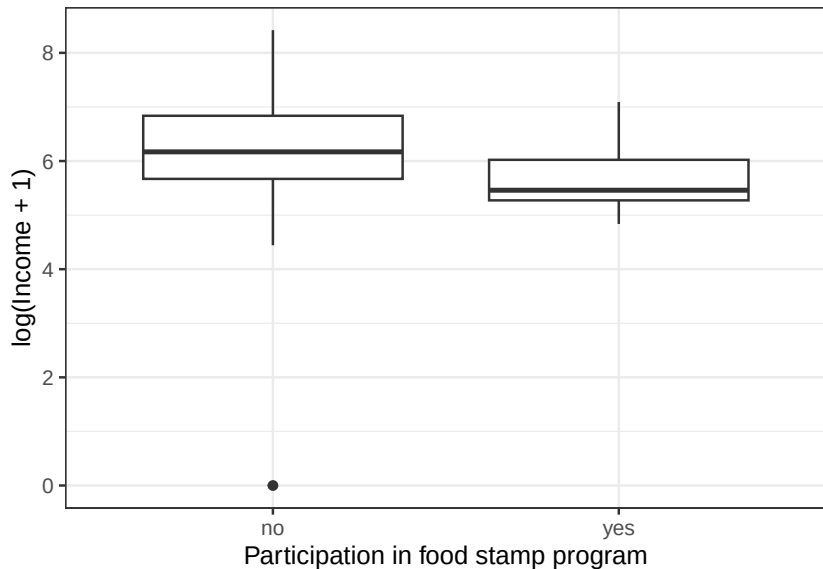
Outliers and multicollinearity

Outliers: Observations which don't follow the same trend as the rest of the data.

Multicollinearity: (linear) relationship between 2 or more explanatory variables.

Both can cause problems for estimating parameters in a regression model.

Example: outliers



Example: outliers

With potential outlier:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.061	1.663	1.239	0.215
log(income + 1)	-0.617	0.278	-2.224	0.026

Without potential outlier:

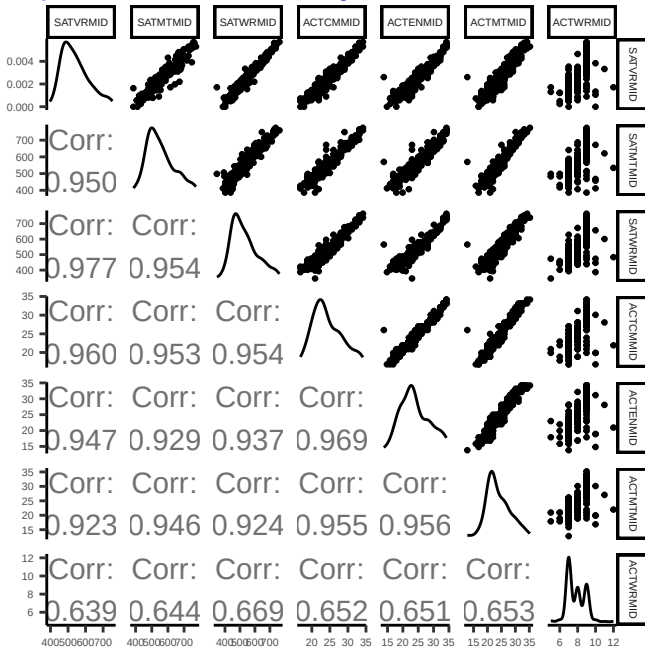
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.961	2.276	2.619	0.009
log(income + 1)	-1.274	0.391	-3.257	0.001

Cook's distance

Handling outliers

Should I just remove any unusual observations in the data?

Example: multicollinearity



Example: multicollinearity

With all the test score variables:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-11.056	1.930	-5.728	0.000
SATVRMID	-0.005	0.011	-0.485	0.628
SATMTMID	0.001	0.008	0.118	0.906
SATWRMID	0.013	0.010	1.209	0.227
ACTCMMID	0.238	0.188	1.266	0.205
ACTENMID	-0.334	0.153	-2.183	0.029
ACTMTMID	0.370	0.154	2.397	0.017
ACTWRMID	0.036	0.203	0.179	0.858

With just one test score variable:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-9.728	1.350	-7.206	0
SATVRMID	0.018	0.003	7.151	0

Variance inflation factors

Handling multicollinearity

```
m1 <- glm(RPY_3YR_70 ~ .,  
          family = binomial, data = scorecard)
```

```
car::vif(m1)
```

```
##    CONTROL  SATVRMID  SATMTMID  SATWRMID  ACTCMMID  ACTEN  
##  3.539098 19.586976 12.138599 18.767795 13.158163 12.693  
##      UGDS  PCTFLOAN      NPT4  
##  2.511964  2.124875  2.438207
```

What should we do when there is high multicollinearity?

Summary

- ▶ Outliers:
 - ▶ Change our coefficient estimates
 - ▶ Identify with EDA before model fitting
 - ▶ Also check with Cook's distance after model fitting
 - ▶ Clear errors, or points from a different population of interest, can be removed. Otherwise, report results with and without outliers
- ▶ Multicollinearity:
 - ▶ Makes it hard to interpret meaning of a coefficient (how do we hold other variables constant??)
 - ▶ Increases variability of estimated coefficients
 - ▶ Identify with pairs plots, correlation matrices before model fitting
 - ▶ Also check with VIFs after model fitting
 - ▶ Options: use a subset of the explanatory variables; combine several explanatory variables (average, PCA, etc.); ignore (if only predicting)